Math 190-1: Mathematics of Sustainability **Spring 2026**

Lecture 7: Dynamic Games

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7.1 Dynamic Games

Example 7.1 *Recall the "hanging with friends" game, where two friends want to hang out but they don't know which venue the other will go to. We can represent it as a table (Tab: 7.1)*

		Player 2	
		A	B
Player 1	A	2,1	0,0
	B	0,0	1,2

Table 7.1: Hanging with Friends as a payoff matrix

However, if we want dynamics, we can represent it as a tree.

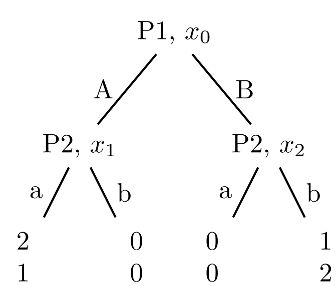
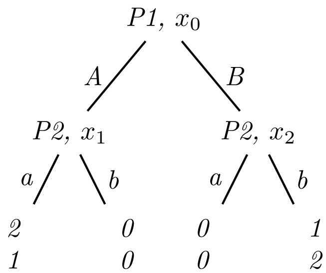
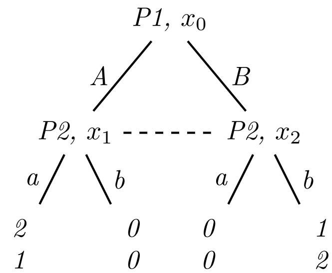


Figure 7.1: Hanging with Friends as a decision tree

Example 7.2 *We need to distingush between the sequential and static "Hanging with Friends" games. For example, we need to make it clear who knows what when. Below you will see two games, with the subtle difference between them. The dashed line between x_1 and x_2 in the second figure means that the information between them cannot be distinguished.*





In the left example, (the one without the dashed line), the information sets we have are $H_1 = \{\{x_0\}\}$ *and* $H_2 = \{\{x_1\}, \{x_2\}\}$. *This means* $H - 2$ *has two sets in them. Since there are two information sets, the player can distingush between the two locations, i.e. player 2 knows what player 1 has done.*

Conversely, the right example has a dashed line between them, so the information sets are $H_1 = \{\{x_0\}\}, H_2 = \{\{x_1, x_2\}\}$. *Since* H_2 *only has one information set, then player 2 cannot distingush between being at* x_1 *or* x_2. *This means that this is a static game.*

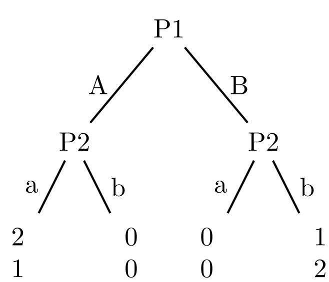
Definition 7.3 *Dynamic games of complete information are called*

- *games of* perfect information *if every information set is a singleton* $\{\{x_1\}\{x_2\}\}$
- *otherwise called* imperfect information, *e.g the information set looks like this:* $\{\{x_1, x_2\}\}$

Definition 7.4 (Extensive form Game) *An* extensive form game

1. *set of players,*
2. *payoff for all players as a function of actions*
3. *order of the moves*
4. *actions of players when they can move*
5. *knowledge of player when they can move (modeled using "information sets")*
6. *Nature can be a player*
7. *structure of the game is common knowledge to all players*

Example 7.5 *Let's start analyzing this game! The strategies for P1 are* A, B, *and the strategies for P2 are deterministic plan of action. For example if P1 chooses* A, *then P2 will choose* a. *If P1 chooses* B, *then P2*



playes b. *But formally, we should choose a strategy for each information set. Since this game is dynamic* *with perfect information, we have* $H_1 = \{\{x_0\}\}, H_2 = \{\{x_1\}, \{x_2\}\}$. *This means the set of actions for P1 are* $s_1(\{x_0\}) = \{A, B\}$ *and the set of actions for P2 are*

$$s_2(\{x_1\}) = \{a, b\}, \qquad s_2(\{x_2\}) = \{a, b\},$$

where the two actions may be different. This means P2 has 4 pure strategies, which we will denote as

$$\{s_2(\{x_1\})s_2(\{x_2\})\} : aa, \quad ab, \quad ba, \quad bb$$

Definition 7.6 (Mixed-strategy) *A* mixed strategy *for player i is a probability distribution over pure strategies.*

Example 7.7 (Mixed Strategy example)

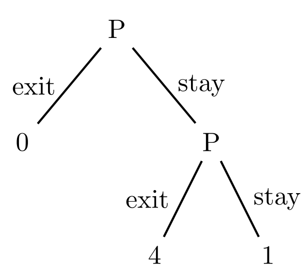
$$\sigma_2 = \begin{cases} \sigma_2(aa) = \frac{1}{10} \\ \sigma_2(ab) = \frac{3}{5} \\ \sigma_2(ba) = \frac{3}{10} \\ \sigma_2(bb) = 0 \end{cases}$$

This means if player 2 is at node x_1, meaning player 1 played A, then player 2 will play a with probability $\frac{1}{10} + \frac{3}{5}$. *And player 2 will play f with probability* $\frac{3}{10} + 0$.

Definition 7.8 *Behavioral Strategy A* behavioral strategy *specifies a probability fistribution over $A_i(h_i)$ for each information set $h_i \in H_i$*

Remark 7.9 *If a player has multiple turns, then these distributions are assumed to be independent, and for good reason.*

Example 7.10 *The concepts of behavioral and mixed strategies aren't the same, as illustrated below. A*



randomized strategy will never recieve a payoff of 4, because player 1 will take the same action in both nodes. However, a behavioral strategy flips a fresh coin at each node.

Behavioral and mixed streategies are the same in perfoect information games.

7.1.1 Strategies and Nash Equilibrium

Example 7.11 (Sequential Hanging with Friends as Normal Form game) *1. Players: $N = 2$*

2. Strategies:

- $s_1 = \{A, B\}$
- $s_2 = \{aa, ab, ba, bb\}$

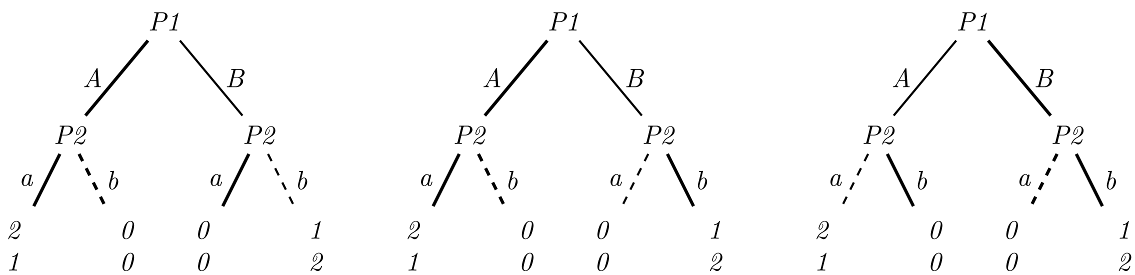
3. *payoffs:*

		Player 2			
		aa	ab	ba	bb
Player 1	A	2,1	2,1	0,0	0,0
	B	0,0	1,2	0,0	1,2

Trying to solve for the pure-strategy Nash Equilibrium, we see

$$\begin{array}{lll} \mathrm{BR}_2(A) = \{aa, ab\} & \mathrm{BR}_1(aa) = \{A\}, & \mathrm{BR}_1(B) = \{A\} \\ \mathrm{BR}_2(B) = \{ab, bb\} & \mathrm{BR}_1(fo) = \{\emptyset\}, & \mathrm{BR}_1(ff) = \{B\} \end{array}$$

This means that the Nash equilbria are: $(A, aa), (A, ab), (B, bb)$.



But are all of these rational? We lost something because of timing. If P1 believes P2 will play bb, then their best response is to play B, but is that rational? If P2 knows they are in x_1*, then their best response should be to play* a.

Similar to if P1 believes P2 is aa, then P1 will play A, but if not, and P1 plays B, then P2's best response should be to play b.